

Access Control Board MQTT Protocol Documentation V1.0.6

Modification history

Version	date	Modified by	Interface involved	Modifications
V1.0.0	2023.08.10	Shi Lei	*	Initialize the interface documentation
V1.0.1	2023.08.18	Shi Lei	Configure interface, online verification, access records	Added sub-device identifier configuration and reporting, added MQTTS configuration, and corrected errors in the online verification extra object.
V1.0.2	2023.08.30	Shi Lei	Alarm reporting	Add configuration card and configuration output code reporting
V1.0.3	2023.08.30	Shi Lei	Configuration Interface	Add fire alarm mode, sub-device version number
V1.0.4	2023.09.14	Shi Lei	Configuration Interface	Add sub-device configuration
V1.0.5	2023.09.05	Ding Hantong	Configuration modification, remote control, firmware upgrade	Adaptation to EC_B10 elevator control board related protocols
V1.0.6	2023.11.27	He Panlong	Configuration Interface	Delete the openMode test mode, change status=0 to test mode, and move the child device information from sysInfo to childInfo.

Applicable Equipment List

Equipment Model	Equipment Name	Compatible Version	Adaptation Date	Remark
CC104	Access control panel	3.1.0	2023-08-10	-
EC_B10	elevator control panel	3.1.0	2023-09-05	-

I. Overview

Application scenarios

This document provides a set of interfaces for direct access to access control devices. After purchasing the equipment, users can directly develop a complete access control or related application system based on the interface specifications. The interface uses the MQTT communication protocol and can interface with various common MQTT brokers; however, we recommend using an access control gateway for unified device management.

Interface Specification

The interface uses the MQTT protocol, and is divided into uplink and downlink in terms of direction:

- ▶ **Uplink** : refers to the device sending a message to the background or application.
- ▶ **Downlink** : refers to the process of sending a message from the background or application to the device.

Based on the initiator, they are divided into: command-type and event-type.

- ▶ **Command-type commands** are initiated by the application platform. First, a **downlink** is sent to the device, followed by an **uplink** , where the device returns the command to the application platform.
- ▶ **Event-based** : This is initiated by the device itself. First , it uploads data to the platform application, then it downloads data **back** to the device , and the platform application sends a response back to the device.

The message format is a standard MQTT message structure, containing a message topic and message content, as follows:

1. Message Topic: The message title, whether for uplink or downlink, the basic rule for sending and returning commands is **the command name** followed by **the command_reply** . The format is as follows:
 - ▶ Downlink: access_device/v1/cmd/{#uuid}/xxxx, corresponding return command (uplink): access_device/v1/cmd/xxxx_reply
 - ▶ Uplink: access_device/v1/event/yyyy, corresponding return command (downlink): access_device/v1/event/{#uuid}/yyyy_reply



Note: `access` is a specific identifier for the access control system; `cmd` indicates that the backend or application actively sends a command to the device; `event` indicates that the device actively sends a message to the backend or application or triggers an event; `{#uuid}` represents the device's unique identifier.

2. Message content : The message content is in text format, and the content is in JSON format.

An example of sending message content is as follows:

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uuid": "0a1b2c3d",
4      "data": {
5          ...
6      },
7      "time": 1647580466,
8      "sign": "e0k4j r i r 85t j e8r u4j r ur 499r 99i i 4ur "
9  }

```

Here is an example of the returned message content:

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uuid": "0a1b2c3d",
4      "code": "000000",
5      "data": {
6          ...
7      },
8      "time": 1647580466,
9      "sign": "e0k4j r i r 85t j e8r u4j r ur 499r 99i i 4ur "
10 }

```

Parameter name	illustrate	type	must	Remark
serialNo	Serial Number	string	yes	When a background application sends a message to a device, it must transmit a unique serial number, not exceeding 32 bits. The device will use the same serial number to provide feedback after receiving the message. Conversely, the same serial number must be transmitted.
uuid	Unique Device Identifier	string	yes	Strings longer than 8 characters
data	Message data body	object	yes	The data is in JSON format, and the message body format varies depending on the message type. The API list mainly details this data format.

Parameter name	illustrate	type	must	Remark
time	Timestamp	long	yes	10-digit timestamp, in seconds
sign	Message Signature	string	no	This value is required if signing is enabled to verify the validity of the data. For signing rules, please refer to: Appendix - Signature Algorithm
code	Result code	string	no	The returned data must contain a code; different codes represent different result identifiers. For basic rules regarding result codes, please refer to: Appendix - Result Codes



Note: Each item in the message content must begin with a lowercase letter in its name, and subsequent names can be a combination of uppercase and lowercase.

II. List of Instruction Interfaces

Configuration Query

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/cmd/{#uuid}/getConfig](#)
- Content: [Standard Format](#)
- The data section of the content:
 - The data field can be empty, or this part can be omitted; in this case, the device will return all configuration data.
 - The data can be a string because the configuration data is a JSON object corresponding to key/value. Therefore, if a specified key is passed in, the value corresponding to that key in the device can be returned.
- Request Example

```

1  //
2  {
3      "serialNo": "6w8kei f5 g6",
4      "uuid": "1b89e112c70940f0 ",
5      "time": 1647580466,
6      "data": "",

```



```

7      "sign": ""
8    }
9
10   //
11   {
12     "serialNo": "6w8kei f5 g6",
13     "uid": "1b89e112c70940f0 ",
14     "time": 1647580466,
15     "data": "net l nfo ",
16     "sign": ""
17   }
18
19   //
20   {
21     "serialNo": "6w8kei f5 g6",
22     "uid": "1b89e112c70940f0 ",
23     "time": 1647580466,
24     "data": "chi l dl nfo ",
25     "sign": ""
26   }

```

► illustrate:

► Flow: Device -> Platform

► Topic: [access_device/v1/cmd/getConfig_reply](#)

► Content: [Standard Format](#)

► The data section is in JSON format, with key/value pairs corresponding to many attributes. See [the attachment for details: Configuration Data Details](#).

It is mainly divided into the following categories:

- **sysInfo** Device system resource information, including device CPU and storage; basic device information, including device type, version, UUID, etc.
 - **netInfo** Device network-related information, including device IP address, Wi-Fi user password, etc.
 - **mqttInfo** MQTT server-side information, including the service's IP address, port, etc.
 - **doorInfo** Information related to the equipment control door, including door opening duration, etc.
 - **childInfo** Device and sub-device configuration information, etc.
- Example of return:

```

1  // net l nfo ,
2  {
3    "serialNo": "6w8kei f5 g6",
4    "uid": "1b89e112c70940f0 ",
5    "code": "000000",
6    "time": 1701061767,
7    "sign": "",
8    "message": "Success",
9    "data": {

```

```

10     "netI nfo ": {
11         "t ype": 1,
12         "dhcp": 1,
13         "i p": "10.102.106.49",
14         "gat eway": "10.102.106.124",
15         "dns": "10.102.106.124",
16         "s ubnet Mask": "255.255.255.0",
17         "net Mac": "66:a7:71:22:c7:09",
18         "nt p": 1,
19         "nt pAddr": "182.92.12.11",
20         "nt pl nt er val ": 43200,
21         "nt pTi meout ": 5,
22         "nt pHour ": 23
23     }
24 }
25 }
26
27 // chi l dl nfo ,
28 {
29     "ser i al No": "6w8kei f5 g6",
30     "uui d": "1b89e112c70940f0 ",
31     "code": "000000",
32     "t i me": 1701061808,
33     "si gn": "",
34     "message": "Success",
35     "data": {
36         "chi l dl nfo ": [
37             {
38                 "i ndex": "01",
39                 "devi cel d": "",
40                 "confi g": "",
41                 "vers i on": ""
42             },
43             {
44                 "i ndex": "02",
45                 "devi cel d": "",
46                 "confi g": "",
47                 "vers i on": ""
48             },
49             {
50                 "i ndex": "04",
51                 "devi cel d": "",
52                 "confi g": "{ post fi x=\"\", w l ed=0, com_pass wd=\"1234567887654321\", haddr =\"
53                 \"vers i on\": \"m20_v11_v3.1.0.6.l hj y.12\"
54             },
55             {
56                 "i ndex": "08",
57                 "devi cel d": "",
58                 "confi g": "",
59                 "vers i on": ""
60         }

```

```
61 | ]
62 | }
63 | }
```

► Special result code

type	code	information	illustrate
Parameter abnormality	200000	No matching data found	The parameter passed may be incorrect; please pay attention to capitalization.

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Configuration modification

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/cmd/{#uuid}/setConfig](#)
- Content: [Standard Format](#)
- The data section of the content:
 - The data field can be empty, or this part can be omitted; in this case, the device will not make any configuration changes.
 - `data` can be a JSON object. Passing in a specified key and value allows you to modify the value corresponding to that key in the device. It supports up to two levels. For example `key1.key2` , `<key>` indicates setting the corresponding data object and then finding the corresponding value `key1` within that object . See [the attachment for details: Configuration Data Details](#) , where some data cannot be modified. `key2`
- Request Example

```
1  //  i p
2  {
3      "serialNo": "6w8kei f5 g6",
4      "uid": "0a1b2c3d",
5      "time": 1647580466,
6      "data": {
7          "netInfo": {
8              "ip": "192.168 .32 .23"
9          }
10     },
11     "sign": ""
12 }
13
14 //
15 {
```

```
16 | "serialNo": "6w8kei f5 g6",
17 | "uid": "0a1b2c3d",
18 | "time": 1647580466,
19 | "data": {
20 |     "childInfo": [{
21 |         "deviceId": "123",
22 |         "config": "___VBAR_CONFIG_V1.1.0___{ devname=\" abc\" } --r /t PDnEg1fL B!
23 |     },
24 |     {
25 |         "deviceId": "456",
26 |         "config": "___VBAR_CONFIG_V1.1.0___{ devname=\" abc\" } --r /t PDnEg1fL B!
27 |     }
28 | },
29 | "sign": ""
30 | }
```

- Flow: Device -> Platform
- illustrate:
- Topic: [access_device/v1/cmd/setConfig_reply](#)
- Content: [Standard Format](#)
- data section: empty
- Example of return:

```
1 | // netInfo.ip,
2 | {
3 |     "serialNo": "6w8kei f5 g6",
4 |     "uid": "0a1b2c3d",
5 |     "time": 1647580466,
6 |     "code": "000000",
7 |     "sign": ""
8 | }
```

- Special result code

type	code	information	illustrate
Parameter abnormality	20000	No matching data found	The parameter passed may be incorrect; please pay attention to capitalization.
Parameter abnormality	200001	The data format to be modified is incorrect.	For example, it should be a String type, but an int type is passed; it can only have two values, 0 and 1, but a 2 is passed.

type	code	information	illustrate
Parameter abnormality	200002	Device configuration refresh failed	The parameter passed may be incorrect; please pay attention to capitalization.
Other abnormalities	300000	This value cannot be modified.	-

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Query permissions

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/cmd/{#uuid}/getPermission](#)
- Content: [Standard Format](#)
- The data section of the content:

Parameter name	illustrate	type	must	Remark
size	Number per page	Int	yes	Maximum of 200 entries per page (only one entry per face type).
page	page number	Int	yes	Which page, starting from 0?
code	voucher number	String	no	voucher number
type	voucher type	Int	no	Refer to Appendix - Device Status Code Table - type column
id	Permission ID	String	no	-

- Request Example

```
1 //
2 {
3     "serialNo": "6w8kei f5 g6",
4     "uuid": "0a1b2c3d",
5     "data": {
6         "page": 0,
7         "size": 200
8     }
9 }
```

```

8      },
9      "time": 1647580466,
10     "sign": ""
11   }
12   // code type
13   {
14     "serialNo": "6w8kei f5 g6",
15     "uid": "0a1b2c3d",
16     "data": {
17       "page": 0,
18       "size": 200,
19       "code": "2ki fk 39k40",
20       "type": 0
21     },
22     "time": 1647580466,
23     "sign": ""
24   }
25   // id, size page
26   {
27     "serialNo": "6w8kei f5 g6",
28     "uid": "0a1b2c3d",
29     "data": {
30       "page": 0,
31       "size": 200,
32       "id": "2ki fk 39k40"
33     },
34     "time": 1647580466,
35     "sign": ""
36   }

```

- illustrate:
- Flow: Device -> Platform
- Topic: [access_device/v1/cmd/getPermission_reply](#)
- Content: [Standard Format](#)
- Data section: JSON object format

Parameter name	illustrate	type	must	Remark
page	Current page	Int	yes	Current page (starting from 0)
size	Page size	Int	yes	Current page size
total	total	Int	yes	Total number of whitelists
totalPage	Total pages	Int	yes	Total pages

Parameter name	illustrate	type	must	Remark
count	Actual number of whitelist entries on the current page	Int	yes	Actual number of whitelist entries on the current page
content	Permission List	Array	yes	Permission List
id	Permission ID	String	yes	This is a unique identifier for permissions; it's used to determine permissions when adding or deleting them. It's recommended to pass a 32-bit random value.
type	Permission types	Int	yes	Refer to Appendix - Device Status Code Table - type column
code	permission value	String	yes	Access permission value
index	Gate number	String	yes	Refer to Appendix - Gated Access Mapping Format . For the gate number, a 1 in the corresponding bit indicates it is valid.
extra	Additional parameters	Object	no	Other additional attributes, such as personnel name, etc.
time	Time range	Object	yes	View time range

► Example of return:

```
1  {
2      "serialNo": "6w8kei f5 g6",
3      "uid": "0a1b2c3d",
4      "code": "000000",
5      "data": {
6          "content": [{
7              "id": "20190101120101",
8              "type": 101,
9              "code": "001002",
10             "index" : "08",
11             "extra": {
12                 },
13             "time": {
14                 "type": 0
15             }
16         }
```

```

17         } ],
18         "page": 1,
19         "size": 20,
20         "total": 100,
21         "totalPage": 5,
22         "count": 20
23     },
24     "time": 1647580466,
25     "sign": ""
    }

```

► Special result code

type	code	information	illustrate
Parameter abnormality	200000	No matching data found	The parameter passed may be incorrect; please pay attention to capitalization.
Parameter abnormality	200001	Exceeding the page limit or having fewer than 0 pages	Exceeding the page limit or having fewer than 0 pages
Parameter abnormality	200002	Exceeding the maximum limit per page	Exceeding 200, or 1 for a face.

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Add permissions

- Flow path: Platform -> Device
- illustrate:
- Topic: [access_device/v1/cmd/{#uuid}/insertPermission](#)
- Content: [Standard Format](#)
- The `data` section of the content is a JSON array object, with each element having the following parameters:

Parameter name	Required	type	illustrate	Remark
id	Permission ID	String	yes	This is a unique identifier for permissions; it's used to determine permissions when adding or deleting them. It's recommended to pass a 32-bit random value.

Parameter name	Required	type	illustrate	Remark
type	Permission types	Int	yes	Refer to Appendix - Device Status Code Table - type column
code	permission value	String	yes	Access permission value
index	Gate number	String	yes	Refer to Appendix - Gated Access Mapping Format . For the gate number, a 1 in the corresponding bit indicates it is valid.
extra	Additional parameters	Object	no	Other additional attributes, such as personnel name, etc.
time	Time range	Object	yes	View time range

► Request Example

```
1  {
2      "serialNo": "6w8kei f5 g6",
3      "uuid": "0a1b2c3d",
4      "data": [{
5          "id": "20190101120101",
6          "type": 101,
7          "code": "001002",
8          "index": "08",
9          "extra": {
10             },
11          "time": {
12              "type": 0
13          }
14      },
15      {
16          "id": "20190101120102",
17          "type": 103,
18          "code": "003004",
19          "index": "01",
20          "extra": {
21             },
22          "time": {
23              "type": 0
24          }
25      }
26  ],
27  "time": 1647580466,
```

```

28 |      "sign": ""
29 |    }

```

- ▶ Note: Success is either all or all will fail; partial success is not possible. If the inserted permission already exists in the device (same ID), it will be ignored, but a success result will still be returned.
- ▶ Flow: Device -> Platform
- ▶ Topic: [access_device/v1/cmd/insertPermission_reply](#)
- ▶ Content: [Standard Format](#)
- ▶ The data section is a JSON array object, where each element is the permission ID for successful insertion.
- ▶ Example of return:

```

1 | {
2 |     "serialNo": "6w8kei f5 g6",
3 |     "uuid": "0a1b2c3d",
4 |     "code": "000000",
5 |     "data": [ "20190101120101", "20190101120102" ],
6 |     "time": 1647580466,
7 |     "sign": ""
8 | }

```

- ▶ Special result code

type	code	information	illustrate
Parameter abnormality	20000	Missing required parameters	Missing required parameters
Parameter abnormality	200001	Permissions cannot be empty.	Permissions cannot be empty.
Parameter abnormality	200002	The number of permissions is greater than a certain number	The number of permissions is greater than a certain number
Insufficient space	200003	Insufficient disk space	Insufficient disk space
Database exception	200004	Database error	Database error
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Deletion Permission

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/cmd/{#uuid}/delPermission](#)
- Content: [Standard Format](#)
- The data section of the content is a JSON array object, where each element is a permission ID.
- Request Example

```

1  //
2  {
3      "serialNo": "6w8kei f5 g6",
4      "uuid": "0a1b2c3d",
5      "data": [ "20190101120101", "20190101120102" ],
6      "time": 1647580466,
7      "sign": ""
8  }
```

- Note: Success is either complete or complete failure; partial success is not possible. If the permission to delete does not exist on the device, it will be ignored, but a success result will still be returned.
- Flow: Device -> Platform
- Topic: [access_device/v1/cmd/delPermission_reply](#)
- Content: [Standard Format](#)
- The data section is a JSON array object, where each element is the permission ID for successful deletion.
- Example of return:

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uuid": "0a1b2c3d",
4      "code": "000000",
5      "data": [ "20190101120101", "20190101120102" ],
6      "time": 1647580466,
7      "sign": ""
8  }
```

- Special result code

type	code	information	illustrate
Parameter abnormality	20000	Missing required parameters	Missing required parameters

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Clear permissions

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/cmd/{#uuid}/clearPermission](#)
- Content: [Standard Format](#)
- The data section of the content: None
- Request Example

```

1  |  //
2  |  {
3  |      "s e r i a l N o": "6w8kei f5 g6",
4  |      "u u i d": "0a1b2c3d",
5  |      "t i m e": 1647580466,
6  |      "s i g n": ""
7  |  }
```

- Note: Failure will not be returned.
- Flow: Device -> Platform
- Topic: [access_device/v1/cmd/clearPermission_reply](#)
- Content: [Standard Format](#)
- Data section: None
- Example of return:

```

1  |  {
2  |      "s e r i a l N o": "6w8kei f5 g6",
3  |      "u u i d": "0a1b2c3d",
4  |      "c o d e": "000000",
5  |      "t i m e": 1647580466,
6  |      "s i g n": ""
7  |  }
```

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Query Key

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/cmd/{#uuid}/getSecurity](#)
- Content: [Standard Format](#)

► The data section of the content:

Parameter name	illustrate	type	must	Remark
size	Number per page	Int	yes	Number of pages per page, maximum 20
page	page number	Int	yes	Which page, starting from 0?
key	key value	String	no	voucher number
type	Key type	String	no	RSA, etc.
id	Key ID	String	no	-

► Request Example

```

1  //
2  {
3      "serialNo": "6w8kei f5 g6",
4      "uid": "0a1b2c3d",
5      "data": {
6          "page": 0,
7          "size": 20
8      },
9      "time": 1647580466,
10     "sign": ""
11 }
12 // key type
13 {
14     "serialNo": "6w8kei f5 g6",
15     "uid": "0a1b2c3d",
16     "data": {
17         "page": 0,
18         "size": 20,
19         "key": "vguang",
20         "type": "RSA"
21     },
22     "time": 1647580466,
23     "sign": ""
24 }
25 // id, size page
26 {
27     "serialNo": "6w8kei f5 g6",
28     "uid": "0a1b2c3d",
29     "data": {
30         "page": 0,
31         "size": 20,
32         "id": "2ki fk 39k40"

```

```

33 |     },
34 |     "time": 1647580466,
35 |     "sign": ""
36 | }

```

- illustrate:
- Flow: Device -> Platform
- Topic: [access_device/v1/cmd/getSecurity_reply](#)
- Content: [Standard Format](#)
- Data section: JSON object format

Parameter name	illustrate	type	must	Remark
page	Current page	Int	yes	Current page (starting from 0)
size	Page size	Int	yes	Current page size
total	total	Int	yes	Total number of whitelists
totalPage	Total pages	Int	yes	Total pages
count	Actual number of whitelist entries on the current page	Int	yes	Actual number of whitelist entries on the current page
content	Permission List	Array	yes	Permission List
id	Permission ID	String	yes	Unique identifier of the key
type	Key type	String	yes	RSA, etc.
key	key value	String	yes	-
value	Key value	String	no	For security reasons, the original value of the key cannot be directly retrieved.
startTime	Start time	Long	no	Timestamp (seconds)
endTime	Expiration time	Long	no	Timestamp (seconds)

- Example of return:

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uuid": "0a1b2c3d",
4      "code": "000000",
5      "data": {
6          "content": [{
7              "type": "RSA",
8              "id": "213v",
9              "key": "vguang",
10             "value": "",
11             "startTime": 1560303425,
12             "endTime": 1683918932
13         }],
14         "page": 1,
15         "size": 20,
16         "total": 100,
17         "totalPage": 5,
18         "count": 20
19     },
20     "time": 1647580466,
21     "sign": ""
22 }

```

► Special result code

type	code	information	illustrate
Parameter abnormality	200000	No matching data found	The parameter passed may be incorrect; please pay attention to capitalization.
Parameter abnormality	200001	Exceeding the page limit or having fewer than 0 pages	Exceeding the page limit or having fewer than 0 pages
Parameter abnormality	200002	Exceeding the maximum limit per page	Exceeding 200, or 1 for a face.

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Add key

- Flow path: Platform -> Device
- Note: Key updates are not supported. You can delete the key and then add it again to update.
- Topic: [access_device/v1/cmd/{#uuid}/insertSecurity](#)
- Content: [Standard Format](#)

- The `data` section of the content is a JSON array object, with each element having the following parameters:

Parameter name	Required	type	illustrate	Remark
id	Permission ID	String	yes	Unique identifier of the key
type	Key type	String	yes	RSA, etc.
key	key value	String	yes	-
value	Key value	String	no	-
startTime	Start time	Long	no	Timestamp (seconds)
endTime	Expiration time	Long	no	Timestamp (seconds)

► Request Example

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uid": "0a1b2c3d",
4      "data": [{
5          "type": "RSA",
6          "id": "213v",
7          "key": "vguang",
8          "value": "MFwwDQYJ KoZI hv cNAQE B BQADS wAwSAJ BAI hGA5XL hPR22MRf7 m 4R3NeI",
9          "startTime": 1560303425,
10         "endTime": 1683918932
11     },
12     {
13         "type": "RSA",
14         "id": "213t",
15         "key": "test",
16         "value": "MFwwDQYJ KoZI hv cNAQE B BQADS wAwSAJ BAI hGA5XL hPR22MRf7 m 4R3NeI",
17         "startTime": 1560303425,
18         "endTime": 1683918932
19     }
20 ],
21     "time": 1647580466,
22     "sign": ""
23 }
```

- Note: Success is either all or all will fail; partial success is not possible. If the inserted key already exists in the device (with the same ID), it will be ignored, but a success result will still be returned.

- Flow: Device -> Platform
- Topic: [access_device/v1/cmd/insertSecurity_reply](#)
- Content: [Standard Format](#)
- The data section is a JSON array object, where each element is the key ID of a successfully inserted key.
- Example of return:

```
1  {
2      "serialNo": "6w8kei f5 g6",
3      "uuid": "0a1b2c3d",
4      "code": "000000",
5      "data": [ "213v", "213t" ],
6      "time": 1647580466,
7      "sign": ""
8  }
```

- Special result code

type	code	information	illustrate
Parameter abnormality	20000	Missing required parameters	Missing required parameters
Parameter abnormality	200001	The key cannot be empty.	The key cannot be empty.
Parameter abnormality	200002	The number of keys is greater than a certain number	The number of keys is greater than a certain number
Insufficient space	200003	Insufficient disk space	Insufficient disk space
Database exception	200004	Database error	Database error
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Delete key

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/cmd/{#uuid}/delSecurity](#)
- Content: [Standard Format](#)
- The data section of the content is a JSON array object, where each element is a permission ID.

► Request Example

```

1  |  //
2  |  {
3  |      "serialNo": "6w8kei f5 g6",
4  |      "uid": "0a1b2c3d",
5  |      "data": [ "213v", "213t" ],
6  |      "time": 1647580466,
7  |      "sign": ""
8  |  }

```

- Note: Success is either complete or complete failure; partial success is not possible. If the permission to delete does not exist on the device, it will be ignored, but a success result will still be returned.
- Flow: Device -> Platform
- Topic: [access_device/v1/cmd/delSecurity_reply](#)
- Content: [Standard Format](#)
- The data section is a JSON array object, where each element is the permission ID for successful deletion.
- Example of return:

```

1  |  { ,
2  |      "uid": "0a1b2c3d",
3  |      "code": "000000",
4  |      "data": [ "213v", "213t" ],
5  |      "time": 1647580466,
6  |      "sign": ""
7  |  }

```

► Special result code

type	code	information	illustrate
Parameter abnormality	20000	Missing required parameters	Missing required parameters

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Clear key

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/cmd/{#uuid}/clearSecurity](#)
- Content: [Standard Format](#)

- The data section of the content: None
- Request Example

```

1  |  //
2  |  {
3  |      "serialNo": "6w8kei f5 g6",
4  |      "uid": "0a1b2c3d",
5  |      "time": 1647580466,
6  |      "sign": ""
7  |  }

```

- Note: Failure will not be returned.
- Flow: Device -> Platform
- Topic: [access_device/v1/cmd/clearSecurity_reply](#)
- Content: [Standard Format](#)
- Data section: None
- Example of return:

```

1  |  {
2  |      "serialNo": "6w8kei f5 g6",
3  |      "uid": "0a1b2c3d",
4  |      "code": "000000",
5  |      "time": 1647580466,
6  |      "sign": ""
7  |  }

```

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Adding faces (not yet implemented)

- Flow path: Platform -> Device
- Note: The system downloads a facial photo to the device and binds it to a code. If multiple photos are downloaded with the same code, the older photo will be overwritten. Only one face can be registered at a time.
- Topic: [access_device/v1/cmd/{#uuid}/insertFace](#)
- Content: [Standard Format](#)

Parameter name	Required	type	illustrate	Remark
code	Personnel ID	String	yes	Unique ID identifier for personnel

Parameter name	Required	type	illustrate	Remark
data	Face base64 or feature value	String	yes	Face base64 or feature value
index	Gate number	String	yes	Refer to Appendix - Gated Access Mapping Format . For the gate number, a 1 in the corresponding bit indicates it is valid.
extra	Additional parameters	Object	no	Other additional attributes, such as personnel name, etc.

► Request Example

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uid": "0a1b2c3d",
4      "data": [{
5          "code": "20190101120101",
6          "data": "ZnNmZHNmYXNkZnFzZGY=. . . . .",
7          "index": "08",
8          "extra": {
9              "name":
10             }
11         },
12         {
13             "code": "20190101120102",
14             "data": "ZnNmZHNmYXNkZnFzZGY=. . . . .",
15             "index": "01",
16             "extra": {
17                 "name":
18             }
19         }
20     ],
21     "time": 1647580466,
22     "sign": ""
23 }
```

► illustrate:

► Flow: Device -> Platform

► Topic: [access_device/v1/cmd/insertFace_reply](#)

► Content: [Standard Format](#)

► Example of return:

```
1  {
2      "serial No": "6w8kei f5 g6",
3      "uui d": "0a1b2c3d",
4      "code": "000000",
5      "time": 1647580466,
6      "sign": ""
7  }
```

► Special result code

type	code	information	illustrate
Parameter abnormality	20000	Missing required parameters	Missing required parameters
Insufficient space	200003	Insufficient disk space	Insufficient disk space
Database exception	200004	Database error	Database error
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Deleting faces (not yet implemented)

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/cmd/{#uuid}/delFace](#)
- Content: [Standard Format](#)
- Request Example

```
1  //
2  {
3      "serial No": "6w8kei f5 g6",
4      "uui d": "0a1b2c3d",
5      "code": "20190101120101",
6      "time": 1647580466,
7      "sign": ""
8  }
```

- illustrate:
- Flow: Device -> Platform
- Topic: [access_device/v1/cmd/delFace_reply](#)
- Content: [Standard Format](#)

- ▶ Example of return:

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uid": "0a1b2c3d",
4      "code": "000000",
5      "time": 1647580466,
6      "sign": ""
7  }

```

- ▶ Special result code

type	code	information	illustrate
Parameter abnormality	20000	Missing required parameters	Missing required parameters

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Clear faces (not yet implemented)

- ▶ illustrate:
- ▶ Flow path: Platform -> Device
- ▶ Topic: [access_device/v1/cmd/{#uuid}/clearFace](#)
- ▶ Content: [Standard Format](#)
- ▶ The data section of the content: None
- ▶ Request Example

```

1  //
2  {
3      "serialNo": "6w8kei f5 g6",
4      "uid": "0a1b2c3d",
5      "time": 1647580466,
6      "sign": ""
7  }

```

- ▶ Note: Failure will not be returned.
- ▶ Flow: Device -> Platform
- ▶ Topic: [access_device/v1/cmd/clearFace_reply](#)
- ▶ Content: [Standard Format](#)
- ▶ Data section: None
- ▶ Example of return:

```

1 | {
2 |     "serialNo": "6w8kei f5 g6",
3 |     "uuid": "0a1b2c3d",
4 |     "code": "000000",
5 |     "time": 1647580466,
6 |     "sign": ""
7 | }

```

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Remote control

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/cmd/{#uuid}/control](#)
- Content: [Standard Format](#)
- The data section of the content contains a JSON object with the following parameters:

Parameter name	illustrate	type	must	Remark
command	Order	Int	yes	0: Restart 1: Remote door opening/direct floor access 2: Device activation 3: Device disabling 4: Device reset 5: Device prompts (UI/voice) 6: Unified control

`command=5` At that time, `extra` the required fields in the JSON were listed in the table below.

Parameter name	illustrate	type	must	Remark
msg	Prompt message	String	no	This message will be displayed in a pop-up window if the device supports a UI.
msgTimeout	Reminder Time	Int	no	Unit: <code>ms</code> , When the device supports a UI, the UI displays <code>msg</code> the time. This item does not exist; the default display time is: <code>1500ms</code>
wavFileName	Audio file name	String	no	If this option does not exist, the default voice will be played. If this option exists, the voice with the corresponding name will be played . For example , if this option exists, the network

Parameter name	illustrate	type	must	Remark
				transmission behavior needs to be disabled. "wavFileName": "9" 9.wav

When controlling the elevator control panel:

command=1 the index field is the floor number;

command=6 the action field in the extra field is a required field, 0 for off and 1 for on.

► Request Example

```

1  //
2  {
3      "serialNo": "6w8kei f5 g6",
4      "uid": "0a1b2c3d",
5      "data": {
6          "command": 0,
7          "extra": {}
8      },
9      "time": 1647580466,
10     "sign": ""
11 }
12 //
13 {
14     "serialNo": "6w8kei f5 g6",
15     "uid": "0a1b2c3d",
16     "data": {
17         "command": 1,
18         "extra": {}
19     },
20     "time": 1647580466,
21     "sign": ""
22 }
23 // UI 9. wav
24 {
25     "serialNo": "6w8kei f5 g6",
26     "uid": "0a1b2c3d",
27     "data": {
28         "command": 5,
29         "extra": {
30             "msg":
31             "msgTimeout": 2000,
32             "wavFileName": "9"
33         }
34     },
35     "time": 1647580466,
36 
```



```

37     "sign": ""
38 }
39
40 //
41 {
42     "serialNo": "6w8kei f5 g6",
43     "uid": "0a1b2c3d",
44     "data": {
45         "command": 1,
46         "index": "0000040000" // 19
47     },
48     "time": 1647580466,
49     "sign": ""
50 }
51 //
52 {
53     "serialNo": "6w8kei f5 g6",
54     "uid": "0a1b2c3d",
55     "data": {
56         "command": 6,
57         "extra": {
58             "action": 1
59         }
60     },
61     "time": 1647580466,
62     "sign": ""
63 }

```

- illustrate:
- Flow: Device -> Platform
- Topic: [access_device/v1/cmd/control_reply](#)
- Content: [Standard Format](#)
- Data section: None
- Example of return:

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uid": "0a1b2c3d",
4      "code": "000000",
5      "time": 1647580466,
6      "sign": ""
7  }

```

- Special result code

type	code	information	illustrate
Parameter abnormality	200000	Missing required parameters	Missing required parameters
Parameter abnormality	200001	Unsupported command values	command value error
Execution exception	200002	Instruction execution failed	Execution failed

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Upgrade firmware

- Note: After receiving the upgrade firmware, the device will first download the firmware file. Once the download is successful, the device will return the result to the application platform and then restart.
- Flow path: Platform -> Device
- Topic: [access_device/v1/cmd/{#uuid}/upgradeFirmware](#)
- Content: [Standard Format](#)
- The data section of the content contains a JSON object with the following parameters:

Parameter name	illustrate	type	must	Remark
type	Upgrade package type	Int	yes	Type value strong check: 0: Local upgrade; 1: Lower-level machine upgrade; 2: Elevator upgrade.
URL	Upgrade package download address, HTTP address	String	yes	When calling the API, access this URL. If the URL is accessible, the device will begin downloading the file; otherwise, the API will return a failure message.
md5	MD5 value of the upgrade package	String	yes	MD5 verification of the upgrade package
extra	Additional parameters	Object	no	Additional information required for lower-level machine upgrades
deviceIndex	Downlink device upgrade information	String	no	This is mandatory for upgrading the lower-level machine. It belongs to the extra object and contains the device number to be upgraded. The corresponding bit is valid when it is 1. That

Parameter name	illustrate	type	must	Remark
				is, 01 corresponds to the device 485_0, and 0f corresponds to the devices 485_0, 485_1, 485_2, and 485_3.
deviceModel	Lower device model	String	no	This is a mandatory data transfer for upgrading the lower-level device; it belongs to the extra object and is used for lower-level device upgrade verification.
deviceVersion	Download device version	String	no	This is required for upgrading the lower-level device; it belongs to the extra object and represents the current version of the lower-level device. It is used for lower-level device upgrade verification.

► Request Example

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uuid": "0a1b2c3d",
4      "data": {
5          "type": 0,
6          "url": "http://host/ai o. tar. xz",
7          "md5": "521c2bdc835d4f1 3b5f9 d6db164f0 881"
8      },
9      "time": 1647580466,
10     "sign": ""
11 }

```

- Note: This only indicates whether the upgrade package download was successful or not. The system will usually restart after downloading the upgrade package, and you will only be able to determine if the upgrade was truly successful after restarting.
- Flow: Device -> Platform
- Topic: [access_device/v1/cmd/upgradeFirmware_reply](#)
- Content: [Standard Format](#)
- Data section: None
- Example of return:

```

1  {
2      "serialNo": "6w8kei f5 g6",
3

```

```
4 |         "uui d": "0a1b2c3d",
5 |         "code": "000000",
6 |         "time": 1647580466,
7 |         "sign": ""
    |     }
```

▸ Special result code

type	code	information	illustrate
Parameter abnormality	200000	Missing required parameters	Missing required parameters
Parameter abnormality	200001	The corresponding firmware download address cannot be downloaded.	The corresponding firmware download address cannot be downloaded.
Parameter abnormality	200002	The corresponding firmware failed MD5 verification after download.	MD5 check failed.
Parameter abnormality	200003	No corresponding equipment	The upgraded device does not exist.

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III. List of Event Class Interfaces

Connection reporting

- Note: Only one report is required.
- Flow: Device -> Platform
- Topic: [access_device/v1/event/connect](#)
- Content: [Standard Format](#)
- The data section of the content contains a JSON object with the following parameters:

Parameter name	illustrate	type	must	Remark
sysVersion	System version number	String	no	System version number
appVersion	Application version number	String	no	Application version number

Parameter name	illustrate	type	must	Remark
createTime	Application release date	Long	no	Application release date
btMac	Bluetooth MAC address	String	yes	Bluetooth MAC address
Mac	Unique Hardware Identifier for Devices	String	no	Unique Hardware Identifier for Devices
clientId	Client ID, which is the same as the device address.	String	yes	Client ID
name	Equipment Name	String	yes	Equipment Name
type	Network type	Int	yes	Network type (1: Ethernet 2: Wi-Fi)
ssid	Wifi name	String	yes	wifi name
pwd	Wi-Fi password	String	yes	WiFi password
dhcp	Is it in DHCP mode?	Int	yes	Is it DHCP (1: dynamic, 0: static)?
IP	IP address	String	yes	IP address
gateway	gateway	String	yes	gateway
DNS	DNS server	String	yes	DNS server
subnetMask	Subnet mask	String	yes	Subnet mask
netMac	MAC address	String	yes	MAC address

► Request Example

```
1 {
2   "serialNo": "6w8kei f5 g6",
3   "uuid": "0a1b2c3d",
4   "data": {
5     "sysVersion": "v1. 0. 3. 4050",
6     "appVersion": "v3. 204. 234. 5",
7     "createTime": 1647550466,
```

```

8      "uui d": "0a1b2c3d",
9      "name": "#12-B 01",
10     "model ": "cc104"
11   },
12   "t i me": 1647580466,
13   "s i gn": ""
14 }

```

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/event/{#uuid}/connect_reply](#)
- Content: [Standard Format](#)
- Data section: None
- Example of return:

```

1  {
2      "s e r i a l No": "6w8kei f5 g6",
3      "uui d": "0a1b2c3d",
4      "code": "000000",
5      "t i me": 1647580466,
6      "s i gn": ""
7  }

```

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Alarm reporting

- Note: There are many types of alarms, and not all devices support all types.
- Flow: Device -> Platform
- Topic: [access_device/v1/event/alarm](#)
- Content: [Standard Format](#)
- The data section of the content contains a JSON object with the following parameters:

Parameter name	illustrate	type	must	Remark
type	State type	Int	yes	Refer to Appendix - Device Status Code Table - type column
value	State value	String	no	Refer to the appendix - Device Status Code Table . If it is "None", no value needs to be transmitted.

Parameter name	illustrate	type	must	Remark
extra	Additional parameters	Object	no	Additional Information
index	Gate number	String	no	Door sensor data is uploaded upon door opening. Refer to Appendix - Door Control Permission Mapping Format for the door control number; a 1 in the corresponding bit indicates validity.
devName	Equipment Name	String	no	Device name of control board
deviceId	Sub-device identification	String	no	The sub-device identifier of the device currently swiping the configuration card.
deviceUuid	Sub-device UUID	String	no	The sub-device UUID of the device currently using the configuration card.
deviceVersion	Sub-device version	String	no	The sub-device version number of the device currently using the configuration card.

► Request Example

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uid": "0a1b2c3d",
4      "data": {
5          "type": 1,
6          "value": "1",
7          "index": "08",
8          "extra": {
9              }
10     },
11     "time": 1647580466,
12     "sign": ""
13 }
14
15 {
16     "serialNo": "6w8kei f5 g6",
17     "uid": "0a1b2c3d",
18     "data": {
19         "type": 202,
20         "index": "08",
21     }

```

```
21 |         "extra": {
22 |             "devName": "abc",
23 |             "deviceId": "123abc",
24 |             "deviceUuid": "123456abc",
25 |             "deviceVersion": "m20_v3.1.0.1"
26 |         }
27 |     },
28 |     "time": 1647580466,
29 |     "sign": ""
30 | }
```

- ▶ illustrate:
- ▶ Flow path: Platform -> Device
- ▶ Topic: [access_device/v1/event/{#uuid}/alarm_reply](#)
- ▶ Content: [Standard Format](#)
- ▶ Data section: None
- ▶ Example of return:

```
1 | {
2 |     "serialNo": "6w8kei f5 g6",
3 |     "uuid": "0a1b2c3d",
4 |     "code": "000000",
5 |     "time": 1647580466,
6 |     "sign": ""
7 | }
```

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Passage Report

- ▶ illustrate:
- ▶ Flow: Device -> Platform
- ▶ Topic: [access_device/v1/event/access](#)
- ▶ Content: [Standard Format](#)
- ▶ The `data` part of the content is a JSON array object with the following parameters:

Parameter name	illustrate	type	must	Remark
id	Permission ID	String	yes	If the permission exists in the device's whitelist, return the corresponding permission ID; otherwise, return "-1".

Parameter name	illustrate	type	must	Remark
code	Access credentials	String	yes	If it's a face, then it's a unique ID for that face.
type	Permission types	Int	yes	Refer to Appendix - Device Status Code Table - type column
index	Gate number	String	yes	Refer to Appendix - Gated Access Mapping Format . For the gate number, a 1 in the corresponding bit indicates it is valid.
time	Passage time	Long	yes	Timestamp unit: seconds
result	Passage Result	Int	yes	1 indicates pass, 0 indicates fail.
error	Reasons for failure	String	no	Failure reason code or specific reason
extra	Additional parameters	Object	no	Other additional attributes, such as the person's name, should be recorded if access fails, including the original data such as the original value of the QR code. {"srcData":"1234567890"}
index	Gate number	String	yes	Refer to Appendix - Gated Access Mapping Format . For the gate number, a 1 in the corresponding bit indicates it is valid.
deviceId	Sub-device identification	String	no	Sub-device identifier with reported QR code

► Request Example

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uid": "0a1b2c3d",
4      "data": [{
5          "id": "20190101120101",
6          "code": "s345463434gg",
7          "type": 101,
8          "time": 1640917147,
9          "result": 0,
10         "index": "01",

```

```

11 |         "extra": {
12 |             "deviceid": "123abc"
13 |         }
14 |     }],
15 |     "time": 1647580466,
16 |     "sign": ""
17 | }

```

- illustrate:
- Flow path: Platform -> Device
- Topic: [access_device/v1/event/{#uuid}/access_reply](#)
- Content: [Standard Format](#)
- Data section: None
- Example of return:

```

1 | {
2 |     "serialNo": "6w8kei f5 g6",
3 |     "uuid": "0a1b2c3d",
4 |     "code": "000000",
5 |     "time": 1647580466,
6 |     "sign": ""
7 | }

```

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Online verification

- illustrate:
- Flow: Device -> Platform
- Topic: [access_device/v1/event/access_online](#)
- Content: [Standard Format](#)
- The `data` section of the content is a JSON object with the following parameters:

Parameter name	illustrate	type	must	Remark
code	Access credentials	String	yes	If it is a human face, then leave it blank.
type	Permission types	Int	yes	Refer to Appendix - Device Status Code Table - type column

Parameter name	illustrate	type	must	Remark
index	Data source	String	yes	Refer to Appendix - Access Control Mapping Format Data Source, indicating which barcode scanner or card reader reported the data from; a corresponding bit of 1 is valid.
time	Passage time	Long	yes	Timestamp unit: seconds
extra	Additional parameters	Object	no	Scalable, typically recording raw data, such as the original text of a scanned QR code.
srcData	Raw data	String	no	Such as the original text of the QR code or the data from a card swipe.
deviceId	Sub-device identification	String	no	Sub-device identifier with reported QR code

► Request Example

```

1  {
2      "serialNo": "6w8kei f5 g6",
3      "uuid": "0a1b2c3d",
4      "data": {
5          "code": "s345463434gg",
6          "type": 103,
7          "time": 1640917147,
8          "index": "01",
9          "extra": {
10             "srcData": "vg: //v103AQAGAHZndWFuZWl AQAAl cOs QMKN9TJ GCK5Afs 14p4or xE+QnI
11             "deviceId": "123abc"
12         }
13     },
14     "time": 1647580466,
15     "sign": ""
16 }

```

► illustrate:

► Flow path: Platform -> Device

► Topic: [access_device/v1/event/{#uuid}/access_online_reply](#)

► Content: [Standard Format](#)

► Data section: None

► Example of return:

```
1 | {
2 |     "serialNo": "6w8kei f5 g6",
3 |     "uuid": "0a1b2c3d",
4 |     "code": "000000", // 300000
5 |     "time": 1647580466,
6 |     "sign": ""
7 | }
```

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IV. Appendix

Result code

type	code	information	illustrate
success	000000	Command executed successfully	none
General error	100000	Unknown error	none
General error	100001	The device has been disabled.	Many commands are not allowed to be executed after the device is disabled.
General error	100002	The device is busy, please try again later.	The device is currently upgrading, setting up the network, or performing other tasks.
General error	100003	Signature verification failed	none
General error	100004	Timeout error	none
General error	100005	Device offline	none
Parameter abnormality	200000-299999	Parameter abnormality	Different commands correspond to different parameter specifications. Please refer to the parameter exception descriptions following each interface.

type	code	information	illustrate
Other abnormalities	300000-399999	Other known causes of abnormality	Different commands correspond to different exceptions. Please refer to the exception descriptions following each interface.

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Equipment alarm type table

type	type	value	index
Door sensor status	0	0: Door sensor closed; 1: Door sensor open	Corresponding door serial number
Fire alarm status	1	0: Fire alarm off, 1: Fire alarm on	none
Unauthorized opening of the door	2	none	Corresponding door serial number
Door opening timeout	3	none	Corresponding door serial number
Door closing timeout	4	none	Corresponding door serial number
Upgrade report	5	0: Upgrade failed; 1: Upgrade completed successfully.	none
Anti-tamper alarm	6	0: Normal; 1: Alarm removed	none
Configure upload code	100	Configuration field table	
M1 Configuration Card	202	No, device information is placed into the extra object.	

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Permission type description

type	num	illustrate
face	300	300: Facial Data
qrcode	100, 101, 103	100: Transparent code, 101: Static code, 103: Dynamic code
password	400	400: Password data, password length: 6~10 characters
card	200	200: Standard card number; 201: CPU encryption card; 202: Sector encryption card
fingerprint	500	500: Fingerprint data
Bluetooth	600	600: Bluetooth door opening certificate
registrationPlate	700	700: License Plate
openButton	800	800: Door opening button

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Time range

The following four time zone settings are supported:

- 1. Default mode, unlimited time, always effective.

```
1 | {
2 |     "type": 0
3 | }
```

- 2. Normal mode, valid from start to end time.

```
1 | {
2 |     "type": 1,
3 |     "range": {
4 |         "beginTime": 1640917147,
5 |         "endTime": 1690917147
6 |     }
7 | }
```

- 3. Daily mode: Valid from a specific time to a specific period each day (timestamp is the number of seconds from midnight to that time).

```
1  {
2      "type": 2,
3      "beginTime": 30600,
4      "endTime": 66600,
5      "range": {
6          "beginTime": 1640917147,
7          "endTime": 1690917147
8      }
9  }
```

- 4. Weekly Recurring Mode: Valid from specific times to specific times each week, Monday through Sunday; invalid at other times (multiple time slots per day do not overlap, supporting a maximum of five slots). The `weekPeriodTime` parameter is as follows: daily time format is "HH:MM-HH:MM", 24-hour time format, e.g., "08:00-09:30|10:00-11:30". A maximum of 5 time slots are allowed. At least one day from 1 to 7 is required; absence of this item indicates no permission. "00:00-24:00" represents valid all day. Multiple time slots are separated by "|" without spaces. A maximum of 5 time slots are allowed per day, e.g., "08:00-09:30|10:00-11:30|12:00-13:30|15:00-16:30|17:00-18:30".

```
1  {
2      "type": 3,
3      "weekPeriodTime": {
4          "1": "9:00-10:00",
5          "2": "12:00-13:30|15:00-16:30"
6      },
7      "range": {
8          "beginTime": 1640917147,
9          "endTime": 1690917147
10     }
11 }
```

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Configuration items

- netInfo** Information about network configuration

Parameter name	illustrate	type	Editable	Remark
type	Network type	Int	yes	Network type (1: Ethernet, 2: Wi-Fi, 4: 4G)

Parameter name	illustrate	type	Editable	Remark
ssid	Wifi name	String	yes	wifi name
pwd	Wi-Fi password	String	yes	WiFi password
dhcp	Is it in DHCP mode?	Int	yes	Is it DHCP (1: dynamic, 0: static)?
IP	IP address	String	yes	IP address
gateway	gateway	String	yes	gateway
DNS	DNS server	String	yes	DNS server
subnetMask	Subnet mask	String	yes	Subnet mask
netMac	MAC address	String	yes	MAC address
NTP	NTP Enable	Int	yes	0: Off; 1: Intermittent synchronization; 2: Scheduled synchronization
ntpAddr	Network Time Server	String	yes	Network Time Server
ntpInterval	Network time synchronization interval	Int	yes	Network time synchronization interval, in seconds, corresponding to NTP=1
ntpHour	Network time synchronization interval	Int	yes	Network time synchronization time: 0~23, corresponding to NTP=2
ntpTimeout	Network time synchronization timeout	Int	yes	Network time synchronization timeout, in seconds

► [mqttInfo](#) Information about MQTT server configuration

Parameter name	illustrate	type	Editable	Remark
mqttAddr	MQTT server IP	String	yes	MQTT address (IP:port; when enabling one-way or two-way authentication in MQTTS, add mqttts:// or ssl:// as the beginning of the address, e.g., ssl://192.168.10.1:8888)

Parameter name	illustrate	type	Editable	Remark
clientId	Client ID	String	yes	Client ID, which defaults to the same as the device address; sending an empty ID "" will restore the default.
mqttName	mqtt username	String	yes	mqtt username
password	MQTT password	String	yes	MQTT password
qos	mqtt qos	int	yes	MQTT qos can take values of 0, 1, or 2.
prefix	mqtt theme prefix	String	yes	mqtt theme prefix
reporttimely	Reporting mode	int	yes	0: Passage priority, 1: Push priority
onlinecheck	Online verification	int	yes	0: No online verification; 1: Offline verification failed, online verification will be attempted.
timeout	Online verification/push priority timeout	int	yes	unit milliseconds
mqttps	Is MQTTS enabled?	int	yes	0: Disabled; 1: SSL enabled, no authentication between parties; 2: One-way authentication; 3: Two-way authentication.

▸ [sysInfo](#) Equipment system resource information body

Parameter name	illustrate	type	Editable	Remark
time	time	Long	yes	time
appVersion	Application version number	String	no	Application version number
releaseTime	Application release date	Long	no	Application release date
Mac	Unique Hardware Identifier for Devices	String	no	Unique Hardware Identifier for Devices

Parameter name	illustrate	type	Editable	Remark
deviceNum	Device number	Int	yes	Device number
deviceName	Equipment Name	String	yes	Equipment Name
model	Equipment type	String	no	Equipment type
status	Basic equipment status	Int	no	0 - Test mode, 1 - Normal, 2 - Disabled, 3 - Fire alarm
disk	Remaining space for equipment	Int	no	The unit for remaining space in the equipment is kb.
memory	Remaining memory space	Int	no	The unit for remaining memory space is KB.
CPU	CPU utilization	Int	no	CPU utilization percentage
accessDevVersion	External device version number	String	no	External device version number
projectName	Project Name	String	yes	Used to distinguish different projects, only needs to be configured once initially.
device0Id	Sub-device identification	String	no	Sub-device identifier, used to distinguish sub-devices
device1Id	Sub-device identification	String	no	Sub-device identifier, used to distinguish sub-devices
device2Id	Sub-device identification	String	no	Sub-device identifier, used to distinguish sub-devices
device3Id	Sub-device identification	String	no	Sub-device identifier, used to distinguish sub-devices

▸ [doorInfo](#) Equipment door opening configuration information

Parameter name	illustrate	type	Editable	Remark
openMode	Equipment door opening modes: 0 - Normal, 1 - Normally Open, 2 - Normally Closed, 3 - Test Mode	Int	yes	Equipment door opening mode: 0 - Normal, 1 - Normally open mode, 2 - Normally closed mode
openTime1	Door 1 opening time, in milliseconds	Int	yes	Door opening time, in milliseconds
openTime2	Door 2 opening time, in milliseconds	Int	yes	Door opening time, in milliseconds
openTime3	Door opening time (milliseconds)	Int	yes	Door opening time, in milliseconds
openTime4	Door opening time (milliseconds)	Int	yes	Door opening time, in milliseconds
outMode	Relay output mode (referring to whether the relay operates on a high-active or low-active state)	Int	yes	0 NO is valid, 1 NC is valid
openTimeout	Door opening timeout	Int	yes	Door opening time, in seconds
reporttimely	Reporting mode	int	yes	0: Passage priority, 1: Push priority
onlinecheck	Online verification	int	yes	0: No online verification; 1: Offline verification failed, online verification will be attempted.
timeout	Online verification/push priority timeout	int	yes	unit milliseconds
relayCtrl	Relay control	Int	yes	0: Single door single control; 1: Double door single control
fireMode	Fire alarm mode	Int	yes	0: Report a road closure 1: Report a road access

- **childInfo** Configuration information of the lower control device

Parameter name	illustrate	type	Editable	Remark
index	Sub-device subscript	String	no	
deviceId	Sub-device identification	String	no	Sub-device identifier, used to distinguish sub-devices (this is required when configuring config).
config	Sub-device configuration	String	yes	The sub-device configuration string must strictly adhere to the device configuration corresponding to the deviceId.
version	Sub-device version	String	no	Sub-device version number

Gated access control mapping format

- Each bit of the data represents the corresponding gate.

Data bytes	Byte bit representation	Byte correspondence gate
1 bit	0: Not effective, 1: Effective	1
2bit	0: Not effective, 1: Effective	2
3bit	0: Not effective, 1: Effective	3
4bit	0: Not effective, 1: Effective	4

For example, 0x08 means that only the fourth door is controlled, while 0x0F means that all doors can be controlled.

- **uiInfo** UI-related configuration information

Parameter name	illustrate	type	Editable	Remark
rotation	Rotation method	Int	yes	0: Landscape mode 0 degrees, 1: 90 degrees, 2: 180 degrees, 3: 270 degrees
statusBar	Bottom status bar	Int	no	0: Turn off the bottom status bar; 1: Turn on the bottom status bar.
horBgImage	Horizontal background image	Base64	yes	Horizontal background image (PNG format only)
verBgImage	Vertical background image	Base64	yes	Vertical background image (PNG format only)

- [bleInfo](#) Bluetooth-related configuration information (currently only supported in the open version)

Parameter name	illustrate	type	Editable	Remark
Mac	MAC address	String	yes	6 bytes (e.g., d0000c05a6dc)
name	Bluetooth name	String	yes	(e.g., f2ecc32fdc)

- [liftInfo](#) Ladder control interface board configuration information

Parameter name	illustrate	type	Editable	Remark
openFloor	Open floors	String	yes	"0000000001"~"FFFFFFFF"("0000070000" (Levels 17, 18, and 19 permissions)) (Currently only supports a maximum of 40 levels)
openTime	Opening Hours	Int	yes	The default value for 1-255 seconds is 10 seconds.
publicFloor	Public floors	String	yes	"0000000001"~"FFFFFFFF" (e.g., "0000070000" (levels 17, 18, and 19 permissions)) (Currently only supports a maximum of 40 levels)
saveConfig	Save configuration	Int	yes	0 (Do not save), 1 (Save)

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